

# Saatvik Palli

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## EDUCATION

### University of Illinois Urbana-Champaign

Master of Computer Science

Champaign, IL

Aug. 2026 – Dec. 2027

### Michigan State University

B.S. in Computer Science, GPA: 3.9/4.0

East Lansing, MI

Aug. 2022 – May 2026

## EXPERIENCE

### Chief Technology Officer

Nexrup Music Discovery ↗

Aug. 2025 – Present

East Lansing, MI

- Enabled week-over-week momentum ranking for **144 independent artists** by building a daily ETL pipeline accumulating 380,000+ datapoints across 10 streaming and social platforms.
- Cut artist leaderboard load times from **3-4 seconds to ~600 ms** by shrinking the initial bundle 57%, replacing  $O(n^2)$  scans with  $O(1)$  hash-set lookups, and deduplicating concurrent API requests behind a TTL cache.
- Defined the data model and service boundaries while mentoring **3 developers and 2 designers** over 9 months through branch-per-developer workflow, code review, and 110+ PRs.

### Data Analytics & Engineering Intern

Auto-Owners Insurance

May 2024 – Aug. 2026

Lansing, MI

- Spearheaded upgrading licensing management to an internal SQL Server system, saving **\$115K over 5 years**, by engineering ETL pipelines and stored procedures that surfaced real-time compliance status.
- Developed a neural network capable of scoring **10,000+ claims per second**, producing real-time difficulty ratings used to balance caseloads across adjusters, with scikit-learn and MLflow.
- Built a Power BI scoring model ranking **500+ adjusters across 43 branches** on **8 weighted claim-handling metrics**, exposing systematic gaps in rep training and technology adoption.
- Analyzed thousands of claims with Pandas and NumPy to project 18-month cost savings of chatbot-assisted salvage handling, generating reports that strengthened vendor negotiations and contract renewal.

### Software Developer

Magna International, Michigan State Capstone

Aug. 2025 – Dec. 2025

East Lansing, MI

- Developed a CAD classification pipeline enabling engineers to find functionally similar parts across **2,000+ automotive components**, by tagging 3D meshes with a vision LLM using extracted geometry statistics.
- Increased tagging accuracy by 64%** through prompt A/B testing and multi-view rendering optimization.
- Cut per-request API cost by ~50%** and eliminated rate-limit failures by moving tagging to an OpenAI Batch API pipeline that chunked 1,800 parts into 90 asynchronous jobs with state persistence for interrupted runs.

### Executive Board

Imagine Software Club at MSU

Aug. 2024 – May 2026

East Lansing, MI

- Interviewed and matched **~50 students to 6 startup projects**, aligning technical experience to effective teams.
- Designed and hosted a Bash scripting workshop for **60+ students** by hosting a custom server environment.

## PROJECTS

### Half Full Calendar ↗ | JavaScript, Electron, Firebase, Google Calendar API

- Built two-way Google Calendar synchronization backed by Firebase persistence, enabling offline access and cross-session state for a desktop calendar since released on the **Mac App Store and Microsoft Store**.
- Contributed frontend fixes to event creation and viewing flows on a team-built Electron application.

### Insurance Risk Factor Neural Network | Python, Flask, Scikit-Learn, Plotly, Git/GitHub

- Led a team of 3 to build a web app visualizing health risk factors and projected financial loss.
- Improved model accuracy by 8%** on insurance dataset by tuning a scikit-learn MLP.

## TECHNICAL SKILLS

**Languages:** Python, SQL, Bash, JavaScript, C++, C, DAX, HTML/CSS

**Technologies:** SQL Server, GCP, Docker, React, Flask, Firebase/Firestore, Pandas, NumPy, PyTorch, Scikit-learn, MLflow, Power BI, Plotly, Git/GitHub, Electron

**Concepts:** Data Modeling, ETL, Data Warehousing, Distributed Systems, Database Systems (SQL & NoSQL), Query Optimization, Algorithms & Data Structures, Machine Learning, Linux

**Interests:** Jazz trumpet, Eagle Scout, Brazilian jiu-jitsu, poker